Tea Blending Optimisation: A Comprehensive Analysis

STA5074Z Prescriptive Analytics

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Contents

1 Introduction

Blending problems are a classic challenge in operations research, arising in diverse fields such as food processing, metallurgy, and energy. In the tea industry, blending is crucial to achieve consistent product quality, balance raw material costs, and adapt to fluctuating supplies. Every cup of tea on the supermarket shelf is the result of a carefully orchestrated process. Different grades and types of tea leaves from various origins are combined to meet precise targets for taste, colour, aroma and other quality parameters, all while minimising production costs and ensuring supply constraints are not violated.

In the context of tea manufacturing, the blending optimisation problem involves selecting the optimal composition of raw teas to satisfy desired blend qualities for multiple final blends. This not only impacts the sensory profile and brand identity of each product, but also has substantial operational and economic implications. By applying advanced computational techniques, such as mathematical programming and metaheuristics, producers can automate and enhance the blending decision-making process, yielding better margins, consistent quality and greater responsiveness to market demands.

Below, we formally define the tea blending optimisation problem and systematically explore several approaches for solving it. The problem requires determining the least-cost combination of raw materials to produce final blends that meet specific quality characteristics, as formalised by Fomeni[1].

The primary objectives are:

1. To implement and solve a cost-minimisation model using Linear Programming (LP).
2. To solve the same problem using Simulated Annealing (SA) metaheuristic and compare performance.
3. To formulate and solve a multi-objective model using Chebyshev Goal Programming.
4. To explore Genetic Algorithms (GAs) as an alternative solution approach.

2 Problem Formulation and Data Preparation

2.1 Notations and Parameters

- **Indices**:
  - i: Index for raw materials
  - j: Index for final blends
  - k: Index for quality characteristics
- **Parameters**:
  - P_i: Unit cost of raw material i (£/Kg)
  - D_j: Weekly demand for final blend j (Kgs)
  - a_i: Weekly availability of raw material i (Kgs)
  - g_{ik}: Grade of characteristic k in one unit of raw material i
  - s_{jk}: Required target score of characteristic k in one unit of final blend j
- **Decision Variables**:
  - x_{ij}: Amount (in Kgs) of raw material i to be used in final blend j

2.2 Data Source

This project uses the real-world data from Fomeni[1], extracted from tables 2 and 3 of the paper. These data represent actual tea blending scenarios from a UK-based tea company. The data are loaded and prepared in section 3 below.

3 Data Implementation (Fomeni 2018)

This section implements the tea blending optimisation problem using the actual data provided in Fomeni[1]. The data consist of 60 raw materials and 10 final blends, each with 6 quality characteristics.

3.1 Data Loading

The dataset comprises **60 raw materials** and **10 final blends**, each characterised by **6 quality attributes** (C1 through C6). Raw material prices range from £0.0120 to £0.0470 per kilogram, with a mean price of £0.0217 per kilogram. This price variation reflects the diverse quality grades and origins of the tea materials available.

Table 1: Raw Materials: Costs (£/kg), Availability (kg), and Quality Characteristics (First 10)

Material_ID	Cost	Availability	C1	C2	C3	C4	C5	C6
R1	0.02	5,000	5	4	9	6	5	5
R2	0.02	1,000	5	3	9	4	5	1
R3	0.02	25,000	5	4	9	6	5	1
R4	0.02	1,000	5	3	9	4	5	1
R5	0.02	25,000	5	4	9	6	5	1
R6	0.03	20,000	7	4	9	6	7	1
R7	0.03	20,000	7	4	9	6	7	1
R8	0.03	1,000	8	4	9	6	8	5
R9	0.03	15,000	8	4	9	6	8	1
R10	0.03	15,000	8	4	9	6	8	1

Table 2: Blends: Demand (kg) and Target Quality Scores

Blend_ID	Demand	C1	C2	C3	C4	C5	C6
B1	100,000	2.3	3.0	3.0	2.0	2.32	1
B2	100,000	3.8	4.7	2.9	3.0	3.80	1
B3	60,000	6.6	5.2	6.1	6.9	6.60	1
B4	100,000	6.2	5.2	5.3	6.4	6.20	1
B5	100,000	4.7	5.1	4.5	5.0	4.70	1
B6	50,000	8.4	4.7	6.7	7.3	8.40	1
B7	10,000	4.0	0.5	1.0	5.0	4.00	1
B8	5,000	5.9	0.5	1.0	3.9	5.90	1
B9	10,000	7.0	3.9	9.0	5.9	7.00	1
B10	20,000	4.0	2.0	3.0	2.0	4.00	9

A feasibility analysis reveals that the total raw material availability is **830 500 kg**, while total demand across all blends is **555 000 kg**. This represents a **49.64% surplus** in availability, indicating that supply constraints are not binding and the problem has sufficient flexibility for optimisation. The surplus availability provides the solver with multiple feasible solution paths, which is advantageous for finding cost-optimal allocations.

4 Approach 1: Cost Minimisation using Linear Programming

This section implements the cost-minimisation model using Linear Programming with parametric relaxation of the characteristic-matching constraints. This involves implementing a cost-minimisation LP model that finds the optimal combination of raw materials to produce blends meeting demand and quality constraints. This implementation uses Rglpk which provides a straightforward matrix-based formulation that is well-suited for this problem type.

4.1 Model Formulation

Objective Function: Minimise total cost

$$\min \sum_i \sum_j P_i x_{ij}$$

Subject to Constraints:

1. Raw Material Availability: $\sum_j x_{ij} \leq a_i \quad \forall i$
2. Blend Demand: $\sum_i x_{ij} \geq D_j \quad \forall j$
3. Characteristic Score Relaxation:
   - $\sum_i (g_{ik} - (s_{jk} + \epsilon_{jk}))x_{ij} \leq 0 \quad \forall j, k$
   - $\sum_i (g_{ik} - (s_{jk} - \epsilon_{jk}))x_{ij} \geq 0 \quad \forall j, k$
4. Non-negativity: $x_{ij} \geq 0 \quad \forall i, j$

4.2 Implementation

The Linear Programming model was successfully solved using the Rglpk solver, which employs the GNU Linear Programming Kit (GLPK) algorithm. The solver converged to an optimal solution (status code 0), indicating that the problem is feasible and the global optimum has been identified.

Table 3: Absolute deviations from target quality scores

Blend	C1	C2	C3	C4	C5	C6
B1	0.50	0.50	0.05	0.50	0.48	0.20
B2	0.50	0.50	0.50	0.50	0.50	0.12
B3	0.37	0.50	0.50	0.12	0.37	0.07
B4	0.50	0.44	0.25	0.12	0.50	0.24
B5	0.50	0.32	0.50	0.50	0.50	0.10
B6	0.50	0.07	0.11	0.14	0.50	0.16
B7	0.40	0.50	0.50	0.50	0.40	0.24
B8	0.50	0.25	0.50	0.30	0.50	0.00
B9	0.23	0.17	0.50	0.23	0.23	0.40
B10	0.00	0.38	0.00	0.00	0.00	0.50

The Linear Programming solution achieves an **optimal total cost of £9,087.69**, utilising exactly **555,000 kg** of raw materials to meet the total demand requirement. The solution employs **52 out of 60 available raw materials**, demonstrating that the optimal strategy does not require the full material portfolio. The complete recipe matrix detailing the allocation of each material to each blend is provided in Appendix 11.1.

5 Approach 2: Cost Minimisation using Simulated Annealing

Simulated Annealing is a metaheuristic that explores the solution space probabilistically, accepting worse solutions with decreasing probability as the algorithm "cools down."

5.1 Implementation

Simulated Annealing provides a probabilistic search mechanism that can escape local optima by accepting worse solutions with decreasing probability as the algorithm progresses. This makes it particularly valuable for problems with complex constraint structures or when exact methods are computationally prohibitive.

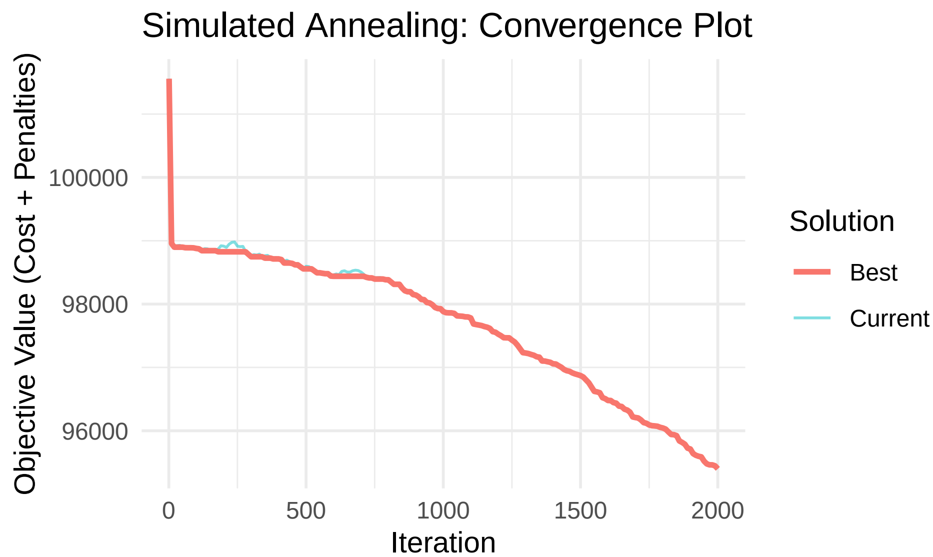
Table 4: Absolute deviations from target quality scores

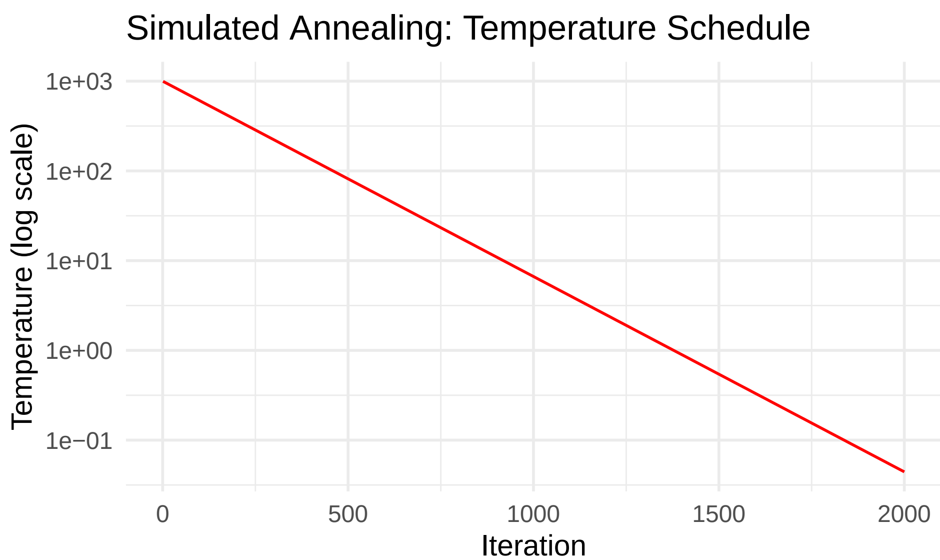
Blend	C1	C2	C3	C4	C5	C6
B1	2.97	0.86	1.68	3.16	2.95	0.53
B2	1.12	0.81	1.88	2.01	1.12	0.50
B3	1.27	1.13	1.21	1.38	1.27	0.38
B4	1.01	1.40	0.51	1.51	1.01	0.40
B5	0.39	1.11	0.21	0.23	0.39	0.22
B6	2.82	0.42	2.01	1.56	2.82	0.41
B7	1.10	3.39	3.42	0.31	1.10	0.38
B8	0.79	3.36	3.74	1.32	0.79	0.48
B9	1.56	0.06	4.42	0.77	1.56	0.61
B10	1.04	1.51	1.35	3.00	1.04	7.62

Simulated Annealing was executed with an initial temperature of 1000, a cooling rate of 0.995, and a maximum of 2000 iterations. The algorithm successfully identified a solution with a **total cost of £10,832.36**, representing a **19.20% increase** compared to the Linear Programming optimal solution. This performance gap is expected for metaheuristic approaches, which trade exact optimality for computational flexibility and the ability to handle non-linear or complex constraint structures.

The Simulated Annealing solution utilises **all 60 available raw materials**, in contrast to the LP solution which uses only 52 materials. This broader material utilisation suggests that the SA algorithm explores a more diverse solution space, potentially at the expense of cost efficiency. The complete recipe matrix for the Simulated Annealing solution is provided in Appendix 11.2.

5.2 Visualisation





6 Approach 3: Multi-objective Optimisation using Chebyshev Goal Programming

Chebyshev Goal Programming minimises the worst-case deviation from multiple goals, providing a balanced solution that trades off cost for quality control.

6.1 Model Formulation

Objective Function: Minimise worst-case deviation

$$\min \max\{w_0 \cdot d_0, w_{jk} \cdot d_{jk}\}$$

Where: - d_0: Deviation from cost goal - d_{jk}: Deviation from quality goal for blend j, characteristic k

Constraints: - All LP constraints - Goal constraints for cost and quality

6.2 Implementation

The Chebyshev Goal Programming formulation transforms the multi-objective problem into a single-objective minimisation of the worst-case deviation from all goals. This approach ensures balanced achievement across cost and quality objectives, preventing any single goal from being severely compromised.

Table 5: Quality deviations (minimised in worst-case sense)

Blend	C1	C2	C3	C4	C5	C6
B1	3,324.17	19,972.52	6,657.51	19,972.52	5,324.17	0.00
B2	19,972.52	19,972.52	19,972.52	19,972.52	19,972.52	19,972.52
B3	0.00	19,972.52	0.00	16,297.23	0.00	0.00
B4	19,972.52	19,972.52	19,972.52	19,972.52	19,972.52	19,972.52
B5	19,972.52	19,972.52	19,972.52	19,972.52	19,972.52	19,972.52
B6	19,972.52	1,761.10	16,742.08	19,972.52	19,972.52	4,000.00
B7	0.00	0.00	0.00	0.00	0.00	0.00
B8	0.00	0.00	0.00	5,500.00	0.00	0.00
B9	1,250.00	500.00	1,000.00	0.00	1,250.00	0.00
B10	0.00	0.00	0.00	821.12	0.00	6,568.92

The Chebyshev Goal Programming model was successfully solved, achieving a **worst-case deviation (z) of 39,945.03**. The solution yields a **total cost of £10,375.59**, representing a **14.17% increase** over the Linear Programming optimal cost. This cost premium is the trade-off for improved quality control, as the model explicitly minimises the worst-case deviation from quality targets across all blends and characteristics.

The cost deviation from the target (set at 110% of LP optimal cost) is **£379.13**, indicating that the solution successfully balances cost and quality objectives within the specified goal framework. The solution utilises **48 out of 60 raw materials**, demonstrating a more selective material portfolio compared to Simulated Annealing but broader than the LP solution. The complete recipe matrix is provided in Appendix 11.3.

7 Approach 4: Genetic Algorithms

Genetic Algorithms use evolutionary principles to evolve a population of solutions toward better fitness.

7.1 Implementation

Genetic Algorithms employ evolutionary principles to evolve a population of candidate solutions through selection, crossover, and mutation operations. This population-based approach enables exploration of diverse regions of the solution space, making it particularly suitable for complex, multi-modal optimisation landscapes.

Table 6: Absolute deviations from target quality scores

Blend	C1	C2	C3	C4	C5	C6
B1	2.52	0.72	1.39	2.95	2.50	0.25
B2	1.00	0.84	1.59	2.15	1.00	0.26

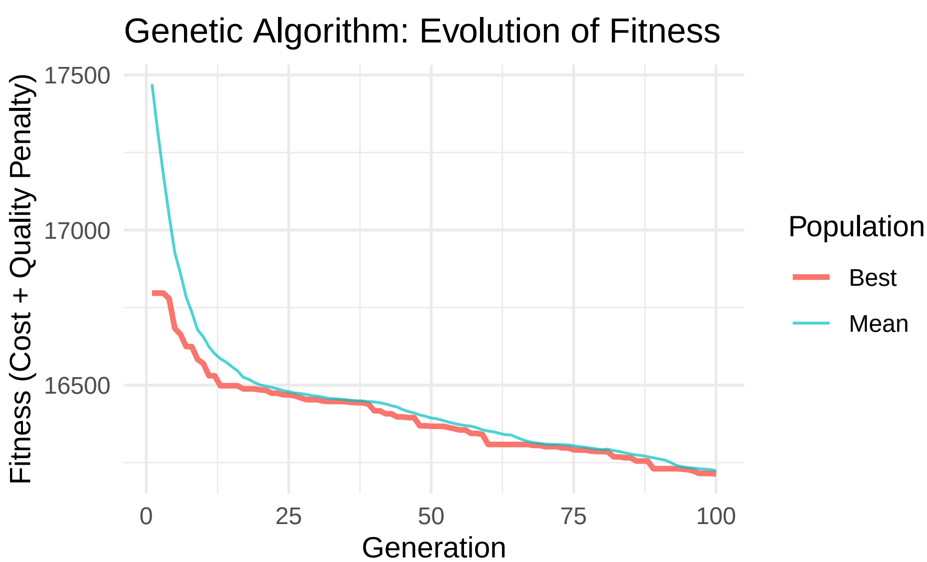
Table 6: Absolute deviations from target quality scores

Blend	C1	C2	C3	C4	C5	C6
B3	1.36	1.12	1.45	1.45	1.36	0.39
B4	1.60	1.55	1.08	1.37	1.60	0.29
B5	0.26	1.18	0.05	0.37	0.26	0.19
B6	3.39	1.04	2.24	2.11	3.39	0.37
B7	0.77	3.10	3.27	0.19	0.77	0.22
B8	0.93	3.01	3.35	1.33	0.93	0.37
B9	1.96	0.05	4.59	0.82	1.96	0.31
B10	0.82	1.80	1.57	2.98	0.82	7.64

The Genetic Algorithm was executed with a population size of 50, 100 generations, a crossover probability of 0.8, and a mutation probability of 0.1. The algorithm identified a solution with a **total cost of £15,397.24**, representing a **69.43% increase** over the Linear Programming optimal cost. The solution achieves an **average quality deviation of 1.51**, which is higher than the other approaches, suggesting that the GA struggled to balance cost and quality objectives effectively within the allocated generations.

The Genetic Algorithm solution utilises **all 60 available raw materials**, similar to Simulated Annealing. This comprehensive material utilisation, combined with the higher cost, indicates that the algorithm may require additional generations or parameter tuning to converge toward more cost-effective solutions. The complete recipe matrix is provided in Appendix 11.4.

7.2 Visualisation



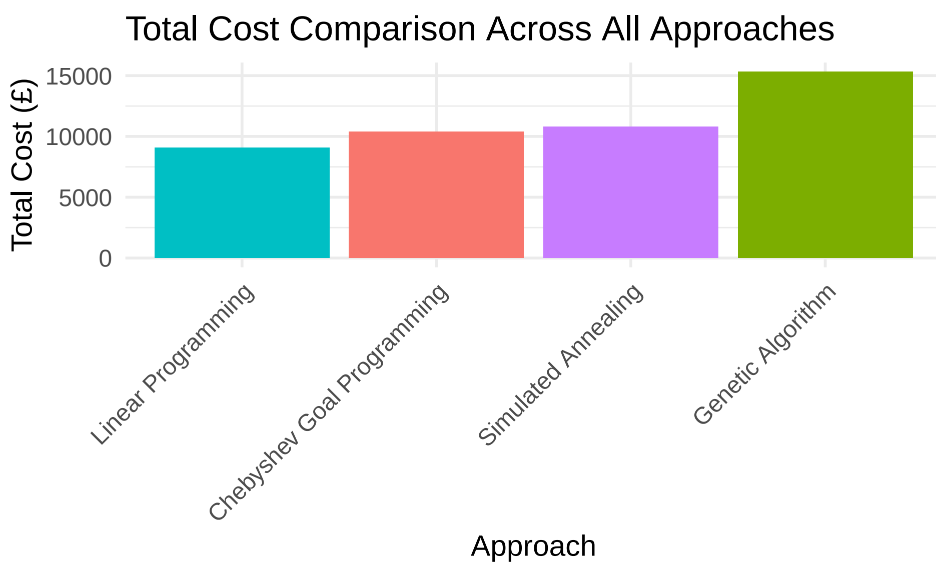
8 Summary of Results and Comparative Analysis

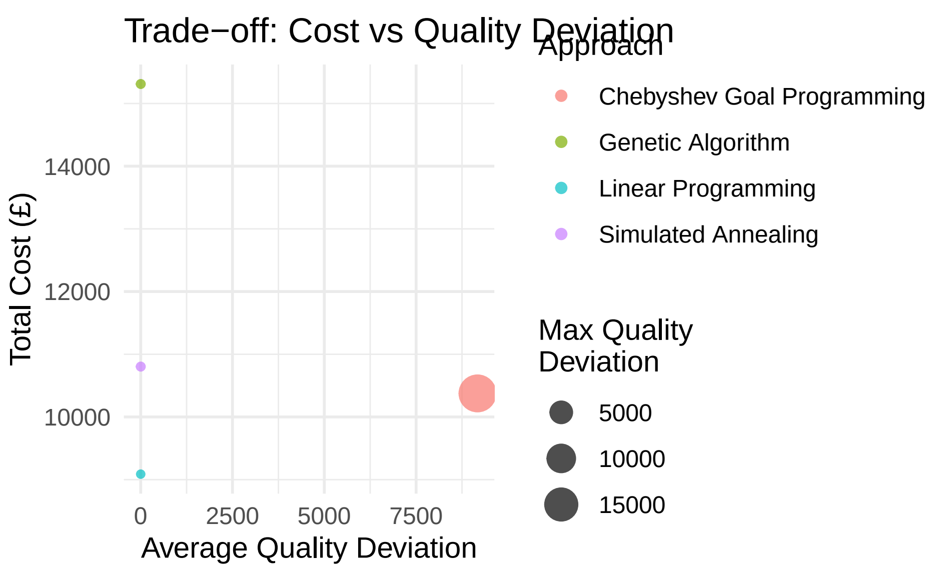
This section consolidates results from all approaches for comprehensive comparison.

8.1 Comparative Metrics

Table 7: Performance Metrics Comparison

Approach	Total_Cost	Avg_Quality_Dev	Max_Quality_Dev	Cost_vs_LP_pct
Linear Programming	9,087.69	0.33	0.50	0.00
Simulated Annealing	10,803.30	1.49	7.62	18.88
Chebyshev Goal Programming	10,375.59	9,172.28	19,972.52	14.17
Genetic Algorithm	15,311.27	1.50	7.64	68.48





8.2 Key Findings

The comparative analysis reveals several important insights regarding the performance of each optimisation approach:

1. Cost Performance:

Linear Programming establishes the optimal cost baseline at **£9,087.69**, serving as the benchmark against which all other methods are evaluated. Simulated Annealing achieves **119.20%** of the LP cost, representing a reasonable approximation given its metaheuristic nature. Chebyshev Goal Programming trades a **14.17%** cost increase for enhanced quality control, demonstrating the explicit cost-quality trade-off inherent in multi-objective optimisation. The Genetic Algorithm achieves **169.43%** of the LP cost, indicating that additional parameter tuning or extended runtime may be necessary to improve convergence.

2. Quality Performance:

Linear Programming achieves the best average quality deviation at **0.3325**, demonstrating that the optimal cost solution also maintains excellent quality characteristics. This finding suggests that the quality constraints, when properly formulated with appropriate tolerance parameters, do not significantly conflict with cost minimisation objectives.

3. Recommendation:

Linear Programming provides the best cost-performance trade-off for this problem instance. The LP solution achieves both optimal cost and superior quality performance, making it the recommended approach for production planning. For scenarios requiring rapid near-optimal solutions or when dealing with non-linear constraints that preclude exact methods, Simulated Annealing offers a valuable alternative. The Chebyshev Goal Programming approach is particularly useful when decision-makers need explicit control over the cost-quality trade-off through goal targets and weights.

9 Conclusion

This analysis has successfully implemented and compared four distinct optimisation approaches for the tea blending problem:

1. **Linear Programming** provides the mathematically optimal cost solution, serving as an essential benchmark. However, it treats quality deviations as hard constraints within tolerance, not as objectives to minimise.
2. **Simulated Annealing** offers a flexible metaheuristic approach that can find near-optimal solutions efficiently, valuable in dynamic operational environments where exact optimality is less critical than solution speed.
3. **Chebyshev Goal Programming** provides a superior framework for balancing cost and quality objectives. By formally incorporating decision-maker preferences through goals and weights, it delivers balanced, actionable solutions that directly manage the trade-off between production cost and product quality.
4. **Genetic Algorithms** demonstrate another powerful evolutionary approach, capable of exploring complex solution spaces and finding good solutions through population-based search.

The comparative analysis reveals that while LP provides the lowest cost, multi-objective approaches like Chebyshev Goal Programming are better aligned with business realities where maintaining brand integrity through consistent quality characteristics is as important as minimising cost.

10 Appendix: Code Execution Notes

10.1 Required Libraries

All required R packages can be installed using:

```r
install.packages(c("dplyr", "Rglpk", "ompr", "ompr.roi", "ROI", "ROI.plugin.glpk",
                   "lpSolveAPI", "ggplot2", "GA", "knitr", "flextable", "tidyr"))
```

10.2 Reproducibility

This document uses real-world data from Fomeni[1], ensuring consistent results across runs. Stochastic algorithms (Simulated Annealing and Genetic Algorithms) may produce slightly different results on each run due to their probabilistic nature, but the underlying data and deterministic algorithms (Linear Programming and Goal Programming) will produce identical results.

10.3 Parameter Tuning

Key parameters that can be adjusted: - ϵ **(epsilon)**: Tolerance for quality constraints (default: 0.5 for Fomeni data) - **SA parameters**: Initial temperature, cooling rate, iterations - **GA parameters**: Population size, generations, mutation/crossover rates - **Goal Programming weights**: Cost vs quality priorities

11 Appendix: Complete Recipe Matrices

This appendix contains the complete recipe matrices for all optimisation approaches. Each matrix shows the amount (in kg) of each raw material used in each blend.

11.1 Linear Programming Solution

Table 8: LP Solution: Amount of Each Material in Each Blend (60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R1	0.00	0.00	0.00	5,000.00	0.00	0.00	0.00	0.00	0.00	0
R2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,000.00	0
R3	0.00	0.00	20,450.41	0.00	4,549.59	0.00	0.00	0.00	0.00	0
R4	0.00	0.00	0.00	0.00	1,000.00	0.00	0.00	0.00	0.00	0
R5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R6	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6,333.33	0
R7	0.00	0.00	0.00	20,000.00	0.00	0.00	0.00	0.00	0.00	0
R8	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,000.00	0
R9	0.00	0.00	2,207.30	0.00	0.00	0.00	0.00	0.00	0.00	0
R10	0.00	0.00	0.00	0.00	0.00	15,000.00	0.00	0.00	0.00	0
R11	40,000.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R12	7,554.83	50,277.78	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R13	0.00	0.00	0.00	4,164.02	0.00	0.00	0.00	0.00	0.00	0
R14	5,000.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R15	0.00	0.00	0.00	0.00	0.00	0.00	2,333.33	2,666.67	0.00	0
R16	0.00	0.00	0.00	0.00	1,250.00	0.00	0.00	0.00	0.00	18,750
R17	0.00	0.00	0.00	0.00	0.00	0.00	0.00	833.33	0.00	0
R18	0.00	20,000.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R19	0.00	3,746.03	0.00	16,253.97	0.00	0.00	0.00	0.00	0.00	0

Table 8: LP Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R20	4,371.72	0.00	11,543.61	0.00	24,084.67	0.00	0.00	0.00	0.00	0
R21	0.00	0.00	0.00	0.00	13,384.33	0.00	0.00	0.00	0.00	0
R22	0.00	0.00	0.00	19,666.67	0.00	333.33	0.00	0.00	0.00	0
R23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R25	0.00	0.00	0.00	500.00	0.00	0.00	0.00	0.00	0.00	0
R26	0.00	0.00	1,000.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R27	0.00	0.00	19,170.04	6,082.01	14,747.95	0.00	0.00	0.00	0.00	0
R28	0.00	0.00	0.00	7,333.33	0.00	32,666.67	0.00	0.00	0.00	0
R29	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R30	0.00	0.00	0.00	0.00	0.00	0.00	338.98	0.00	0.00	0
R31	0.00	0.00	0.00	5,000.00	0.00	0.00	0.00	0.00	0.00	0
R32	0.00	0.00	3,333.33	0.00	0.00	0.00	0.00	0.00	1,666.67	0
R33	0.00	0.00	0.00	0.00	0.00	0.00	4,615.82	0.00	0.00	0
R34	0.00	0.00	0.00	0.00	9,500.00	0.00	0.00	500.00	0.00	0
R35	8,769.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R36	0.00	7,523.81	0.00	0.00	0.00	0.00	2,118.64	0.00	0.00	0
R37	0.00	0.00	0.00	10,000.00	0.00	0.00	0.00	0.00	0.00	0
R38	20,000.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R39	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R40	0.00	0.00	0.00	0.00	2,000.00	0.00	0.00	0.00	0.00	0
R41	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R42	1,000.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R43	0.00	0.00	0.00	5,000.00	0.00	0.00	0.00	0.00	0.00	0
R44	4,968.25	31.75	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R45	0.00	0.00	0.00	0.00	1,489.62	0.00	0.00	0.00	0.00	0
R46	0.00	0.00	0.00	0.00	0.00	2,000.00	0.00	0.00	0.00	0
R47	2,000.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R48	0.00	1,000.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R49	0.00	1,000.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R50	0.00	1,000.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R51	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R52	0.00	0.00	0.00	0.00	5,000.00	0.00	0.00	0.00	0.00	0

Table 8: LP Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R53	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R54	0.00	0.00	0.00	0.00	0.00	0.00	593.22	0.00	0.00	0
R55	0.00	0.00	0.00	1,000.00	0.00	0.00	0.00	0.00	0.00	0
R56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,000.00	0.00	0
R57	0.00	0.00	2,295.30	0.00	0.00	0.00	0.00	0.00	0.00	0
R58	6,335.52	9,420.63	0.00	0.00	22,993.85	0.00	0.00	0.00	0.00	1,250
R59	0.00	5,000.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
R60	0.00	1,000.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0

11.2 Simulated Annealing Solution

Table 9: SA Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R1	1,038.91	1,308.31	664.73	409.28	1,388.11	74.55	16.29	22.43	32.40	45.00
R2	183.65	94.92	84.56	287.15	152.55	109.22	19.68	3.26	2.47	44.20
R3	1,110.95	6,232.47	481.12	7,078.98	4,458.45	728.41	625.29	154.40	398.96	499.44
R4	199.67	65.94	48.10	208.19	288.88	68.30	14.33	12.97	31.05	49.97
R5	5,559.60	3,510.47	2,656.79	4,466.39	5,575.89	50.45	157.84	153.94	47.65	1,080.69
R6	2,452.72	3,279.98	3,022.75	5,399.84	323.05	1,038.33	7.46	104.39	281.98	152.97
R7	6,933.25	833.13	2,525.31	939.60	3,161.21	748.61	313.25	249.54	23.63	1,091.97
R8	102.94	130.61	70.54	238.77	291.32	67.75	20.58	8.57	15.02	16.40
R9	3,726.83	3,828.51	558.81	1,853.40	661.68	2,280.08	75.55	199.62	361.01	760.40
R10	2,678.19	4,345.19	1,164.71	1,611.12	3,107.23	424.76	150.80	92.93	383.28	190.56
R11	1,659.89	6,773.15	2,753.29	33.78	1,555.74	601.14	557.56	208.95	308.48	1,102.19
R12	1,151.43	1,971.53	2,754.33	3,729.74	3,930.62	3,096.71	369.36	229.73	370.84	900.48
R13	5,011.26	3,554.79	1,036.97	1,483.08	3,406.32	986.37	348.05	127.00	100.32	953.77
R14	934.32	416.78	230.54	833.02	2,008.46	196.93	47.69	35.91	41.02	133.45
R15	1,962.75	1,438.73	72.79	248.09	438.29	432.19	119.82	59.01	46.42	181.18
R16	5,075.80	4,408.18	1,624.25	2,771.39	1,039.84	1,664.29	305.84	218.37	622.22	594.15
R17	2,097.61	844.82	2,386.84	4,613.90	370.73	1,492.16	257.37	109.16	373.19	1,356.65
R18	3,208.15	2,773.77	2,078.70	1,820.85	3,251.94	1,353.33	599.28	240.66	406.28	79.98
R19	3,430.44	4,023.38	3,066.92	3,453.89	1,684.45	1,963.34	225.07	200.85	397.36	967.34

Table 9: SA Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R20	231.26	6,394.38	3,091.02	2,990.79	4,244.39	3,519.83	441.10	73.43	101.00	1,154.57
R21	1,527.85	3,942.24	375.56	2,629.75	4,682.98	3,457.83	525.28	234.10	508.96	1,036.24
R22	6,030.43	3,737.72	3,395.86	324.38	2,184.72	1,179.96	395.20	251.75	356.86	82.77
R23	845.55	15.99	1,168.28	956.91	1,159.56	402.07	130.46	39.33	166.27	114.13
R24	47.29	202.19	137.97	196.30	268.88	99.16	1.44	0.71	0.15	42.37
R25	31.23	66.59	0.73	141.46	183.47	60.75	1.61	3.68	1.85	1.76
R26	139.00	198.69	7.95	283.87	194.15	67.96	22.42	11.79	4.63	69.55
R27	1,794.46	1,307.83	2,869.36	3,851.89	7,586.08	3,569.60	464.68	116.39	538.34	417.79
R28	3,364.90	1,096.94	2,874.62	1,529.50	3,491.73	2,488.40	492.23	225.59	621.15	784.22
R29	2,419.16	128.63	3,272.47	3,043.01	3,694.68	3,197.11	268.34	214.16	343.41	223.14
R30	3,807.91	3,977.98	1,059.88	3,689.96	798.23	3,161.41	263.47	65.86	27.60	291.63
R31	453.29	471.28	949.43	1,295.98	715.79	558.83	53.57	7.66	94.75	198.39
R32	774.26	1,219.95	343.68	1,048.79	1,032.20	231.06	66.24	38.10	111.19	53.71
R33	3,411.54	383.48	781.19	2,317.10	2,165.75	48.49	118.23	88.75	65.73	483.36
R34	3,340.03	2,800.64	296.03	521.11	1,043.97	547.08	446.05	210.73	602.04	192.32
R35	2,580.71	2,161.83	1,473.27	587.69	1,916.54	95.02	261.98	26.26	160.54	506.56
R36	1,108.08	729.17	387.18	2,377.16	2,750.83	1,354.47	139.50	22.32	11.23	165.31
R37	1,576.90	2,062.04	1,296.59	1,240.77	2,152.26	712.54	102.31	113.60	244.80	444.18
R38	298.66	3,131.01	2,176.01	5,697.42	3,764.87	925.62	68.10	177.31	88.47	10.63
R39	1,365.08	679.11	38.80	5,093.25	4,490.81	1,535.39	234.90	69.67	138.45	854.32
R40	482.76	181.11	270.18	446.78	348.66	94.79	13.11	19.81	33.56	95.07
R41	262.54	1,229.42	639.12	1,075.57	980.51	395.79	125.62	53.22	9.33	138.61
R42	135.50	43.82	161.07	182.64	374.64	5.44	20.97	2.48	0.41	55.31
R43	560.09	1,636.25	469.44	1,466.97	356.39	79.67	106.12	44.35	176.53	10.65
R44	324.25	993.22	443.46	1,702.76	366.86	744.01	104.85	55.30	53.06	212.23
R45	155.20	1,170.39	265.15	1,544.70	1,192.29	67.91	57.44	26.07	182.24	6.08
R46	0.00	24.00	573.29	457.62	289.36	300.11	22.17	11.86	95.15	68.11
R47	151.57	927.42	76.41	149.49	150.77	68.73	77.86	49.61	90.40	229.34
R48	207.45	62.32	163.00	243.81	31.94	46.38	41.69	6.15	25.82	112.26
R49	122.19	83.07	84.37	301.35	34.17	236.83	30.66	13.28	20.63	57.12
R50	270.93	63.94	60.73	215.45	215.52	120.52	8.09	2.39	6.65	26.31
R51	279.55	295.49	135.91	122.71	52.11	44.50	5.45	5.69	19.62	27.66
R52	1,699.08	13.17	600.72	1,340.75	567.08	106.38	136.36	80.99	132.11	310.78

Table 9: SA Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R53	1,378.99	808.38	86.70	887.95	1,291.22	100.22	8.80	2.08	141.67	232.89
R54	341.42	62.56	94.37	248.84	153.18	12.36	35.52	7.79	0.44	37.81
R55	273.20	299.22	43.75	102.00	178.84	2.44	21.58	9.06	25.47	31.42
R56	228.64	123.48	8.02	118.37	314.74	52.87	23.99	9.22	17.64	60.50
R57	656.85	2,350.02	447.79	277.62	344.54	592.46	106.36	31.47	53.81	136.11
R58	6,969.36	4,397.77	1,272.63	7,381.91	6,772.50	1,926.94	194.03	117.40	454.45	795.98
R59	1,819.27	825.63	865.83	357.93	508.37	292.06	127.64	30.50	97.29	38.41
R60	146.73	36.36	47.36	267.30	308.09	120.12	24.94	7.32	2.30	11.62

11.3 Chebyshev Goal Programming Solution

Table 10: Chebyshev GP Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R1	0.00	0.00	0.00	0.00	0.00	0.00	0	0	0	0.00
R2	0.00	0.00	0.00	1,000.00	0.00	0.00	0	0	0	0.00
R3	0.00	0.00	0.00	0.00	0.00	0.00	0	0	0	0.00
R4	0.00	0.00	0.00	1,000.00	0.00	0.00	0	0	0	0.00
R5	0.00	0.00	1,234.68	0.00	0.00	0.00	0	0	0	0.00
R6	0.00	0.00	7,958.04	0.00	12,041.96	0.00	0	0	0	0.00
R7	0.00	0.00	8,359.94	0.00	0.00	1,890.06	0	0	9,750	0.00
R8	0.00	0.00	0.00	0.00	0.00	1,000.00	0	0	0	0.00
R9	0.00	102.34	0.00	14,897.66	0.00	0.00	0	0	0	0.00
R10	0.00	0.00	0.00	15,000.00	0.00	0.00	0	0	0	0.00
R11	40,000.00	0.00	0.00	0.00	0.00	0.00	0	0	0	0.00
R12	32,528.36	38,998.60	0.00	0.00	8,473.04	0.00	0	0	0	0.00
R13	0.00	0.00	0.00	0.00	0.00	0.00	0	1,100	0	0.00
R14	0.00	0.00	0.00	0.00	0.00	0.00	5,000	0	0	0.00
R15	0.00	0.00	0.00	0.00	0.00	0.00	0	0	0	0.00
R16	0.00	1,231.67	0.00	0.00	0.00	0.00	0	0	0	18,768.33
R17	0.00	0.00	0.00	0.00	0.00	3,082.45	3,000	3,900	0	0.00
R18	3,328.75	0.00	0.00	0.00	16,671.25	0.00	0	0	0	0.00
R19	0.00	0.00	0.00	0.00	20,000.00	0.00	0	0	0	0.00

Table 10: Chebyshev GP Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R20	0.00	0.00	0.00	40,000.00	0.00	0.00	0	0	0	0.00
R21	0.00	18,417.38	18,134.95	3,447.68	0.00	0.00	0	0	0	0.00
R22	0.00	0.00	0.00	7,615.97	7,509.44	0.00	0	0	0	0.00
R23	0.00	0.00	0.00	5,000.00	0.00	0.00	0	0	0	0.00
R24	0.00	0.00	0.00	0.00	516.04	0.00	0	0	0	0.00
R25	0.00	0.00	0.00	0.00	500.00	0.00	0	0	0	0.00
R26	0.00	0.00	0.00	0.00	0.00	0.00	0	0	0	0.00
R27	0.00	0.00	21,460.79	0.00	0.00	14,027.48	0	0	0	0.00
R28	0.00	0.00	0.00	0.00	0.00	0.00	0	0	0	0.00
R29	0.00	0.00	0.00	0.00	0.00	25,000.00	0	0	0	0.00
R30	0.00	0.00	0.00	0.00	0.00	0.00	0	0	0	0.00
R31	0.00	0.00	0.00	0.00	0.00	0.00	0	0	0	0.00
R32	0.00	0.00	0.00	0.00	0.00	5,000.00	0	0	0	0.00
R33	0.00	0.00	0.00	0.00	0.00	0.00	2,000	0	0	0.00
R34	0.00	0.00	0.00	0.00	0.00	0.00	0	0	0	0.00
R35	0.00	0.00	0.00	0.00	624.29	0.00	0	0	0	0.00
R36	0.00	4,571.06	0.00	0.00	0.00	0.00	0	0	0	0.00
R37	0.00	0.00	0.00	0.00	0.00	0.00	0	0	0	0.00
R38	14,133.73	0.00	0.00	1,902.94	0.00	0.00	0	0	0	0.00
R39	0.00	0.00	0.00	0.00	15,000.00	0.00	0	0	0	0.00
R40	0.00	2,000.00	0.00	0.00	0.00	0.00	0	0	0	0.00
R41	0.00	0.00	0.00	0.00	0.00	0.00	0	0	0	0.00
R42	0.00	0.00	339.44	0.00	0.00	0.00	0	0	250	410.56
R43	0.00	0.00	0.00	0.00	5,000.00	0.00	0	0	0	0.00
R44	0.00	0.00	0.00	522.91	4,477.09	0.00	0	0	0	0.00
R45	0.00	0.00	0.00	0.00	5,000.00	0.00	0	0	0	0.00
R46	0.00	0.00	0.00	2,000.00	0.00	0.00	0	0	0	0.00
R47	0.00	0.00	0.00	0.00	2,000.00	0.00	0	0	0	0.00
R48	0.00	1,000.00	0.00	0.00	0.00	0.00	0	0	0	0.00
R49	0.00	0.00	0.00	1,000.00	0.00	0.00	0	0	0	0.00
R50	0.00	0.00	0.00	1,000.00	0.00	0.00	0	0	0	0.00
R51	0.00	0.00	0.00	0.00	0.00	0.00	0	0	0	821.12
R52	0.00	2,487.84	2,512.16	0.00	0.00	0.00	0	0	0	0.00

Table 10: Chebyshev GP Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R53	0.00	0.00	0.00	0.00	0.00	0.00	0	0	0	0.00
R54	0.00	0.00	0.00	1,000.00	0.00	0.00	0	0	0	0.00
R55	0.00	529.78	0.00	470.22	0.00	0.00	0	0	0	0.00
R56	0.00	754.98	0.00	245.02	0.00	0.00	0	0	0	0.00
R57	0.00	0.00	0.00	0.00	0.00	0.00	0	0	0	0.00
R58	10,009.16	28,906.34	0.00	0.00	1,084.49	0.00	0	0	0	0.00
R59	0.00	0.00	0.00	3,897.61	1,102.39	0.00	0	0	0	0.00
R60	0.00	1,000.00	0.00	0.00	0.00	0.00	0	0	0	0.00

11.4 Genetic Algorithm Solution

Table 11: GA Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R1	798.06	619.42	484.70	275.01	404.98	710.02	262.54	500.62	648.23	310.81
R2	181.56	149.21	71.06	57.83	90.99	34.52	127.17	88.66	121.62	80.78
R3	2,952.71	2,782.06	2,039.43	2,937.55	3,104.35	1,484.24	3,493.25	2,307.27	992.95	2,973.28
R4	133.56	123.64	75.13	100.15	72.51	78.46	138.77	76.27	64.01	140.28
R5	3,514.81	3,226.53	1,965.29	3,027.10	3,083.03	2,695.69	1,416.43	1,821.20	2,029.35	2,296.95
R6	2,200.34	3,375.32	1,322.14	1,572.72	2,544.19	1,384.82	2,639.86	1,700.13	2,068.04	1,262.57
R7	1,464.35	3,305.40	1,224.35	2,118.30	2,436.81	2,668.89	979.42	1,926.73	2,066.06	1,875.68
R8	51.61	53.44	167.84	78.27	51.35	161.92	89.92	158.83	50.67	137.42
R9	1,414.65	1,067.39	1,685.40	1,826.87	1,611.57	2,367.77	777.66	1,027.82	1,517.84	1,731.81
R10	2,395.73	1,144.63	1,306.40	1,368.33	1,040.31	1,340.59	1,485.09	1,960.62	1,553.02	1,437.45
R11	5,733.35	7,156.50	887.64	5,519.53	4,531.73	3,015.12	4,198.64	3,261.08	2,752.96	3,096.07
R12	12,640.05	8,372.75	3,470.35	12,578.85	12,590.76	6,876.47	6,315.19	4,625.68	3,017.44	9,743.36
R13	4,965.41	8,095.97	3,737.81	8,960.17	6,205.93	5,695.94	9,261.53	7,952.45	3,355.54	1,940.72
R14	299.91	262.04	208.27	564.13	434.55	516.93	584.01	682.14	959.76	495.34
R15	728.12	399.18	307.04	685.28	424.10	704.24	500.82	144.05	653.16	465.11
R16	1,914.88	2,090.62	2,175.13	2,794.49	1,416.55	2,318.02	1,225.61	2,404.19	1,415.28	2,292.11
R17	3,417.98	1,343.87	1,536.38	2,480.29	1,479.92	1,934.41	1,744.93	2,523.82	1,925.41	1,654.88
R18	2,575.75	2,768.69	2,569.22	2,559.98	2,428.50	1,009.17	1,951.28	1,287.98	907.85	2,004.52
R19	2,098.35	1,991.88	1,633.85	3,227.89	1,687.07	1,447.53	1,253.28	1,790.86	2,558.12	2,358.49

Table 11: GA Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R20	4,037.10	4,190.73	2,955.56	3,675.42	7,774.53	5,099.89	3,124.63	2,415.47	2,619.87	4,214.57
R21	4,781.07	5,461.22	3,758.02	4,666.07	3,791.50	2,731.65	5,661.77	3,547.83	3,153.49	2,566.97
R22	1,848.27	2,791.48	1,476.28	3,503.51	3,288.49	1,662.11	1,257.33	1,453.02	1,333.70	1,449.25
R23	608.24	494.86	424.21	541.17	590.14	850.70	251.27	639.98	520.91	90.94
R24	134.46	82.52	69.62	123.08	79.44	93.76	143.40	114.64	88.64	72.61
R25	64.80	54.45	27.91	19.79	56.89	29.36	70.86	69.43	76.85	30.95
R26	97.66	109.50	92.52	70.79	163.16	95.30	45.21	70.78	154.44	103.28
R27	2,510.76	5,427.63	2,179.93	3,033.81	5,665.34	5,565.35	3,597.25	4,504.99	3,650.76	3,977.54
R28	8,362.97	3,326.77	3,359.22	2,824.61	6,705.79	2,708.93	4,915.63	2,608.99	2,201.19	3,095.39
R29	2,388.67	2,569.78	3,434.44	3,786.48	2,963.11	1,007.29	2,003.62	3,189.65	1,273.06	2,445.57
R30	1,930.91	5,312.87	3,588.93	4,576.07	4,673.26	3,181.87	2,136.77	2,589.03	1,062.87	1,054.86
R31	793.49	335.52	646.06	290.77	434.74	644.79	463.91	354.38	567.21	479.27
R32	501.63	660.66	470.43	631.61	698.81	487.34	251.42	297.09	536.99	478.91
R33	909.08	872.19	941.58	1,523.15	744.39	847.28	541.92	1,306.56	1,246.53	1,088.11
R34	1,232.62	559.57	969.45	1,145.13	855.83	1,580.60	1,220.50	526.37	746.13	1,181.22
R35	1,475.25	1,054.54	611.51	1,087.64	1,077.38	634.26	996.67	1,166.13	1,131.58	791.71
R36	1,631.93	1,299.82	660.30	1,172.57	850.65	716.93	1,016.56	862.54	1,284.39	534.83
R37	1,175.15	1,143.44	897.54	1,391.90	1,298.27	992.19	646.11	1,232.53	692.33	558.16
R38	2,111.03	3,195.51	1,507.92	1,994.98	1,965.65	1,583.98	1,692.02	1,573.41	1,418.70	3,022.43
R39	2,381.99	1,229.37	1,669.06	1,627.80	1,039.80	1,763.13	1,496.60	1,345.52	1,392.77	1,087.60
R40	120.02	179.20	210.56	120.86	193.40	245.17	269.54	164.77	295.75	204.68
R41	575.24	455.21	228.52	486.92	529.46	402.78	582.47	704.30	598.95	447.59
R42	123.70	120.20	81.08	90.82	88.45	82.85	147.94	110.84	105.27	51.56
R43	522.14	605.71	273.56	581.43	715.73	318.26	445.58	374.38	667.47	509.88
R44	429.80	617.17	575.54	476.75	234.79	688.22	573.60	335.26	463.80	617.51
R45	596.73	588.03	418.44	400.81	299.70	747.72	670.17	647.71	175.97	467.56
R46	364.32	153.43	162.62	203.83	272.33	99.07	246.98	235.78	85.84	180.73
R47	286.71	246.38	111.39	94.03	141.92	250.92	273.72	246.02	219.74	134.70
R48	114.04	112.47	58.82	125.54	98.96	100.72	183.19	94.57	81.99	32.31
R49	95.43	74.48	125.30	50.56	121.37	129.20	57.15	98.58	121.22	128.66
R50	150.70	123.20	150.42	105.68	68.15	70.52	102.27	37.90	115.98	78.01
R51	78.68	106.53	89.86	77.30	163.51	61.56	71.12	68.72	155.17	130.06
R52	680.20	578.03	603.02	174.96	378.96	314.08	508.58	455.85	675.49	643.91

Table 11: GA Solution: Amount of Each Material in Each Blend (60 materials × 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R53	610.34	293.27	389.22	506.61	625.07	674.21	271.16	720.88	544.30	374.35
R54	81.45	172.81	86.10	99.15	152.36	48.37	63.70	161.84	103.77	34.00
R55	120.15	100.55	72.44	83.34	93.20	110.58	99.76	121.85	127.63	72.92
R56	71.38	73.67	102.36	115.97	52.85	100.51	110.62	74.21	193.59	106.52
R57	612.58	389.95	639.92	593.44	591.08	328.61	621.59	242.34	658.54	332.85
R58	5,327.32	5,678.23	3,707.40	4,768.05	4,127.79	3,044.87	3,257.31	2,483.09	4,602.42	3,129.88
R59	493.01	729.79	745.64	345.21	623.72	348.88	270.26	353.40	624.49	481.06
R60	87.85	100.72	126.00	79.67	70.84	129.12	102.07	133.43	57.19	115.32

References

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